Hospital Data Analysis Using SQL

Project Documentation

Project Title: Hospital Data Analysis Using SQL

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Role: Aspiring Data Analyst

Tools Used: MySQL Workbench, SQL, Microsoft Word

Database: MySQL

Table Name: hospital_data

Dataset: Hospital_Data.csv

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1. Introduction

Hospitals generate a significant amount of operational data every day, including patient admissions, doctor availability, hospital departments, discharge details, and medical expenses. Analyzing this data helps healthcare organizations improve resource allocation, monitor patient trends, evaluate departmental performance, and manage operational costs.

This project demonstrates how SQL can be used to analyze hospital data by answering business-oriented questions using aggregate functions, grouping, sorting, and date functions.

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2. Project Objectives

The objectives of this project are:

- Calculate the total number of patients across hospitals.
- Analyze the average number of doctors available in each hospital.
- Identify departments with the highest and lowest patient counts.
- Determine the hospital with the highest medical expenses.
- Calculate daily average medical expenses.
- Analyze patient hospital stay durations.
- Generate city-wise patient reports.
- Generate department-wise average stay reports.
- Generate monthly medical expense reports.
- Practice SQL queries used in real-world healthcare data analysis.

3. Tools and Technologies

Tool	Purpose
MySQL Workbench	Database Management
SQL	Data Querying and Analysis
Microsoft Word	Documentation
PDF	Final Project Submission

4. Dataset Description

The Hospital_Data.csv dataset contains hospital-related information used for SQL analysis.

Column Name	Description
Hospital Name	Name of the hospital
Location	City where the hospital is located
Department	Medical department
Doctors Count	Number of doctors available
Patients Count	Number of patients treated
Admission Date	Date of patient admission
Discharge Date	Date of patient discharge
Medical Expenses	Total medical expenses

5. Database Creation

```
CREATE DATABASE hospital_analysis;


USE hospital_analysis;
```

6. Importing Dataset

The Hospital_Data.csv file was imported into MySQL Workbench using the **Table Data Import Wizard**. After successful import, the dataset was stored in the **hospital_data** table for analysis.

7. SQL Analysis

Query 1: Total Number of Patients

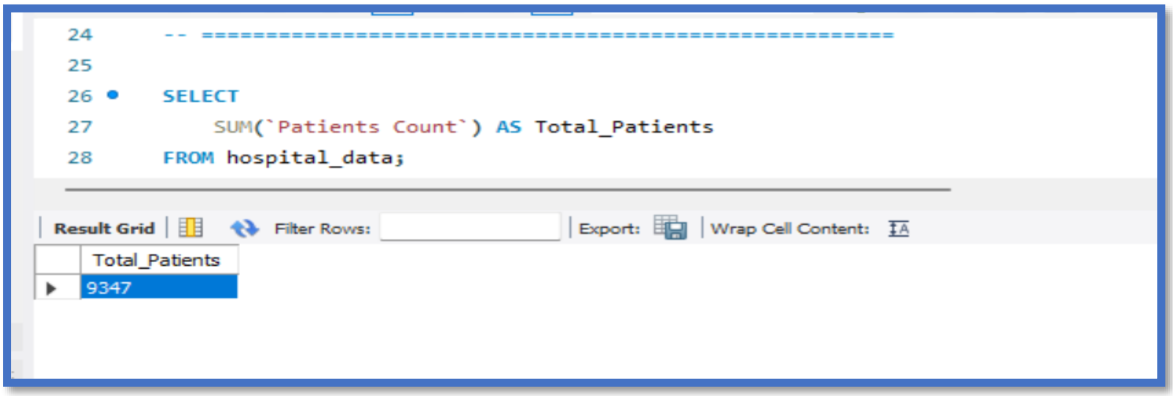
Objective

Calculate the total number of patients across all hospitals.

SQL Query

```
SELECT
    SUM(`Patients Count`) AS Total_Patients
FROM hospital_data;
```

Output



Observation

The query calculates the total number of patients by summing all values in the **Patients Count** column.

Query 2: Average Number of Doctors per Hospital

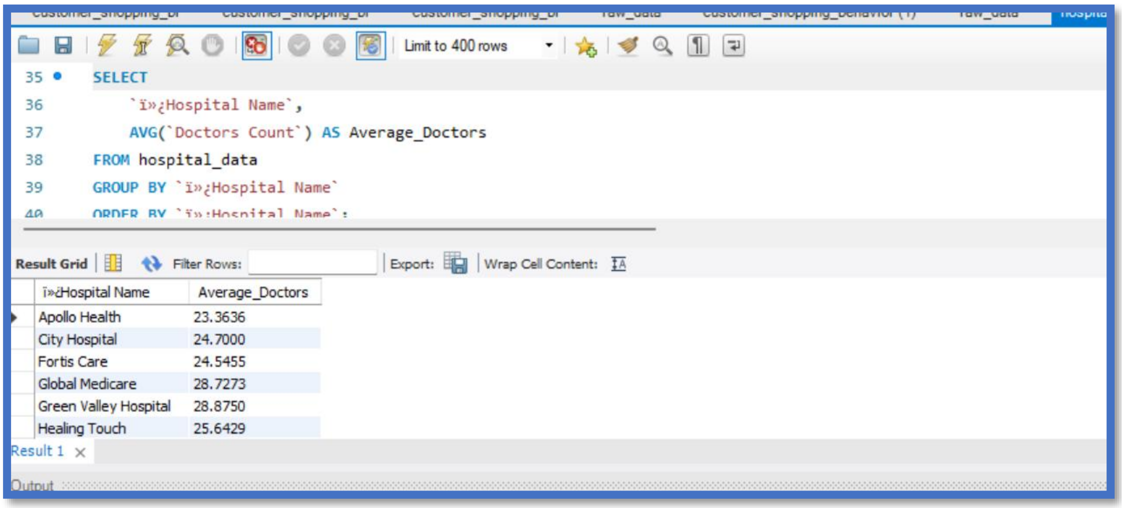
Objective

Calculate the average number of doctors available in each hospital.

SQL Query

```
SELECT
    `Hospital Name`,
    AVG(`Doctors Count`) AS Average_Doctors
FROM hospital_data
GROUP BY `Hospital Name`;
```

Output



Observation

The query groups the records by hospital and calculates the average number of doctors available in each hospital.

Query 3: Top 3 Departments with Highest Number of Patients

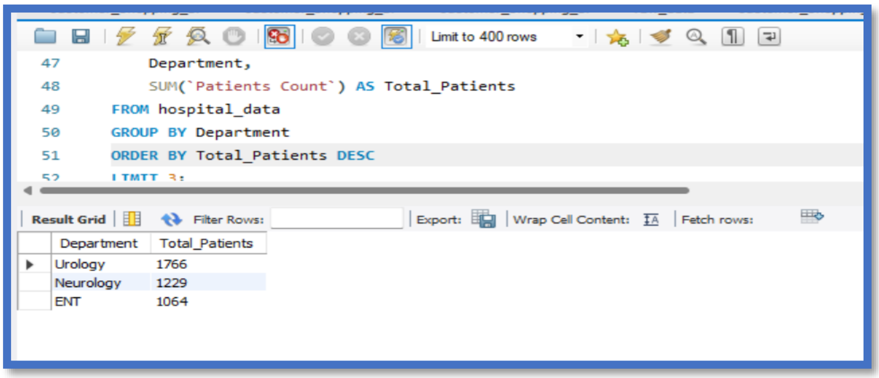
Objective

Identify the three departments treating the highest number of patients.

SQL Query

```
SELECT
    Department,
    SUM(`Patients Count`) AS Total_Patients
FROM hospital_data
GROUP BY Department
ORDER BY Total_Patients DESC
LIMIT 3;
```

Output



Observation

The query identifies the top three departments based on the total number of patients treated.

Query 4: Hospital with Maximum Medical Expenses

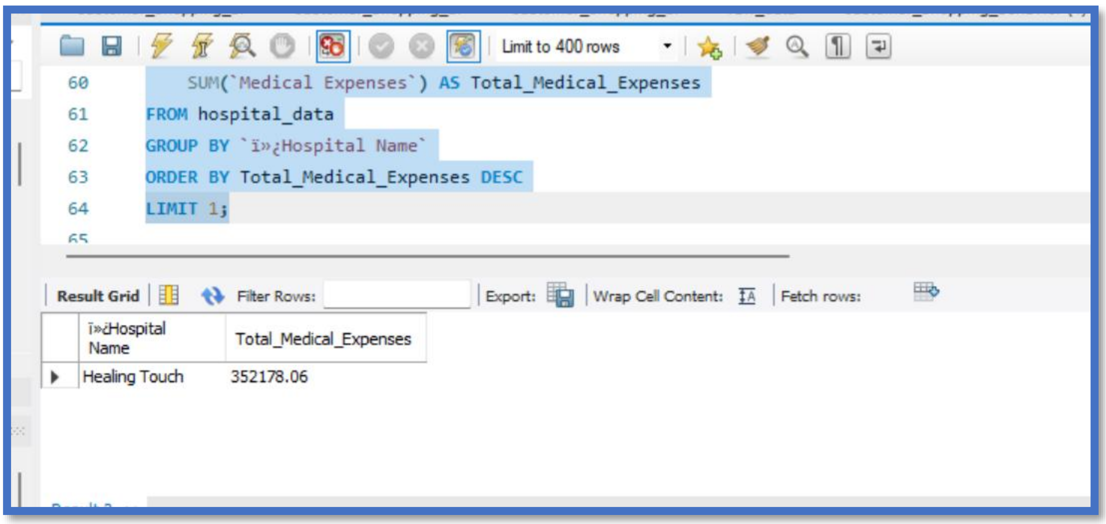
Objective

Find the hospital with the highest total medical expenses.

SQL Query

```
SELECT
    `Hospital Name`,
    SUM(`Medical Expenses`) AS Total_Medical_Expenses
FROM hospital_data
GROUP BY `Hospital Name`
ORDER BY Total_Medical_Expenses DESC
LIMIT 1;
```

Output



Observation

The query identifies the hospital that recorded the highest medical expenses.

Query 5: Daily Average Medical Expenses

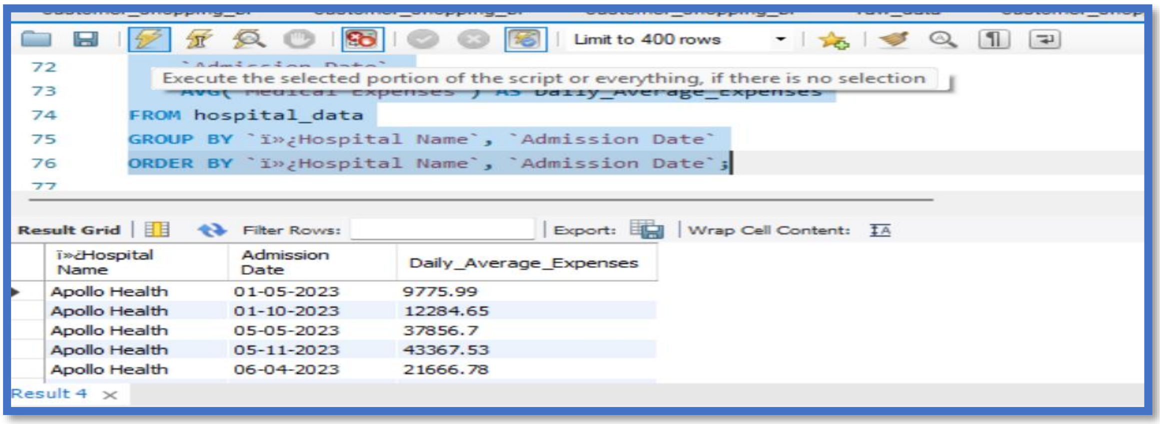
Objective

Calculate the average medical expenses per day for each hospital.

SQL Query

```
SELECT
    `ï»¿Hospital Name`,
    `Admission Date`,
    AVG(`Medical Expenses`) AS Daily_Average_Expenses
FROM hospital_data
GROUP BY `Hospital Name`, `Admission Date`;
```

Output



Observation

The query calculates the average medical expenses for each hospital on every admission date.

Query 6: Longest Hospital Stay

Objective

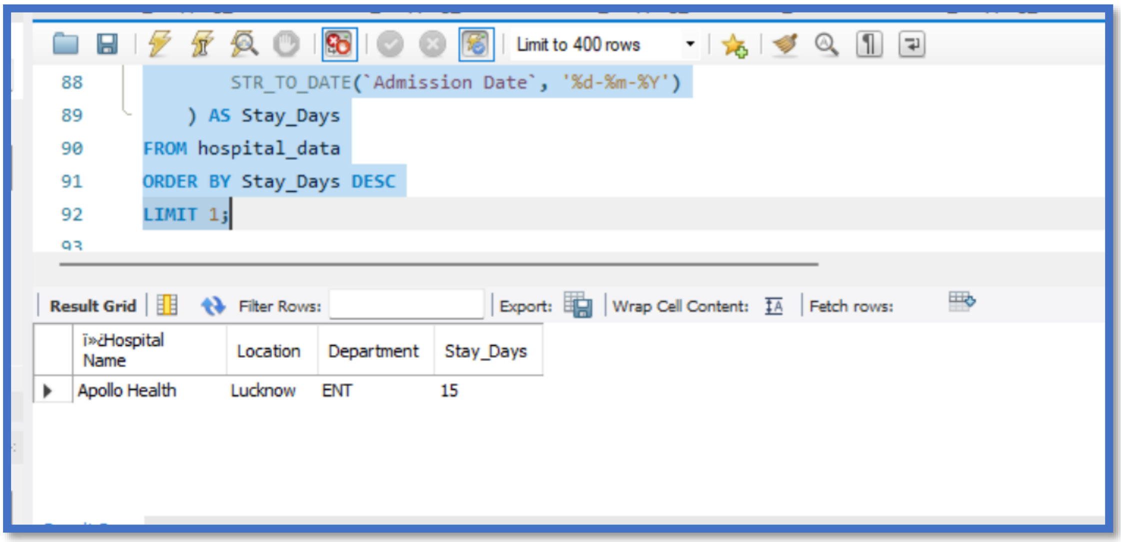
Find the record with the longest hospital stay.

SQL Query

```
SELECT
    `Hospital Name`,
```

```
    Department,
    DATEDIFF(
        STR_TO_DATE(`Discharge Date`,'%d-%m-%Y'),
        STR_TO_DATE(`Admission Date`,'%d-%m-%Y')
    ) AS Stay_Days
FROM hospital_data
ORDER BY Stay_Days DESC
LIMIT 1;
```

Output



Observation

The query calculates the difference between discharge and admission dates to determine the longest hospital stay.

Query 7: Total Patients Treated Per City

Objective

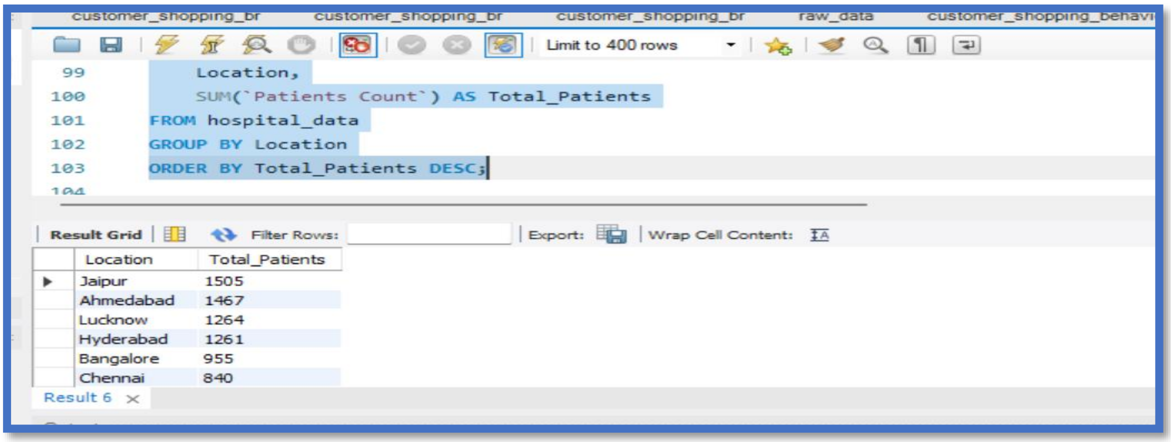
Calculate the total number of patients treated in each city.

SQL Query

```
SELECT
    Location,
    SUM(`Patients Count`) AS Total_Patients
FROM hospital_data
```

GROUP BY Location;

Output



Observation

The query summarizes the total number of patients treated in each city.

Query 8: Average Length of Stay Per Department

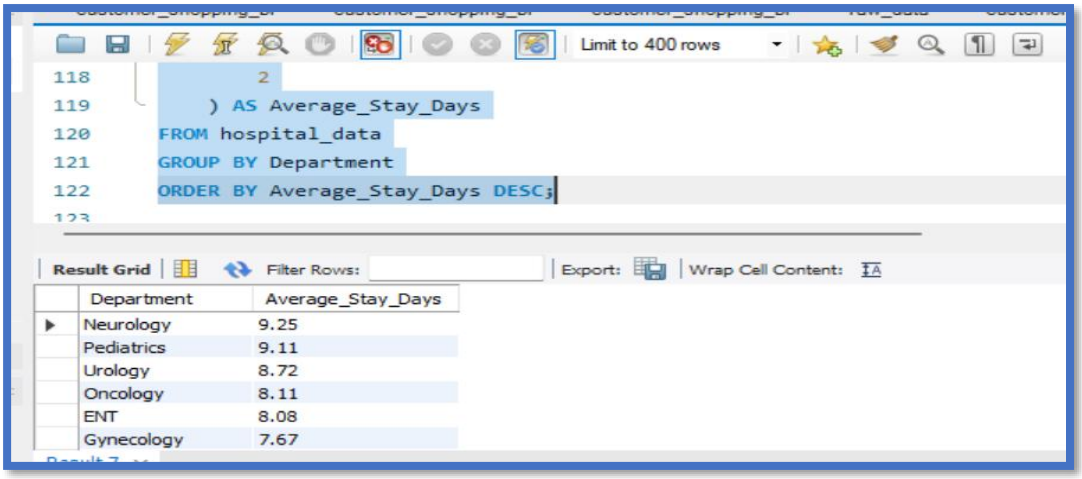
Objective

Calculate the average hospital stay for each department.

SQL Query

```
SELECT
   Department,
   AVG(
      DATEDIFF(
         STR_TO_DATE(`Discharge Date`,'%d-%m-%Y'),
         STR_TO_DATE(`Admission Date`,'%d-%m-%Y')
      )
   ) AS Average_Stay_Days
FROM hospital_data
GROUP BY Department;
```

Output



Observation

The query calculates the average number of days patients remain admitted in each department.

Query 9: Department with Lowest Number of Patients

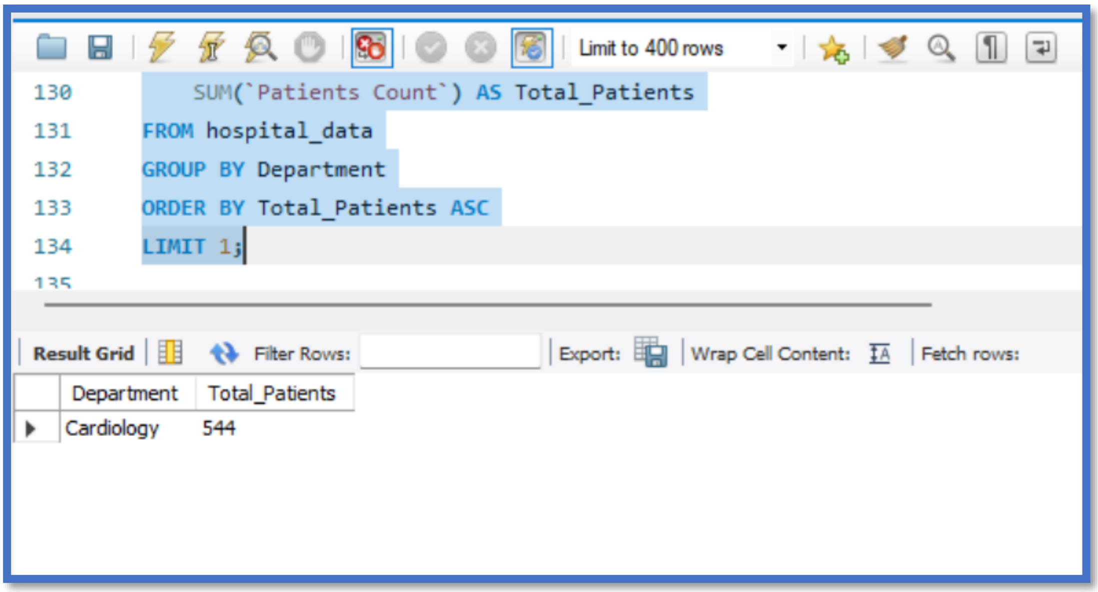
Objective

Identify the department with the lowest number of patients.

SQL Query

```
SELECT
    Department,
    SUM(`Patients Count`) AS Total_Patients
FROM hospital_data
GROUP BY Department
ORDER BY Total_Patients ASC
LIMIT 1;
```

Output



Observation

The query identifies the department with the fewest patients in the dataset.

Query 10: Monthly Medical Expenses Report

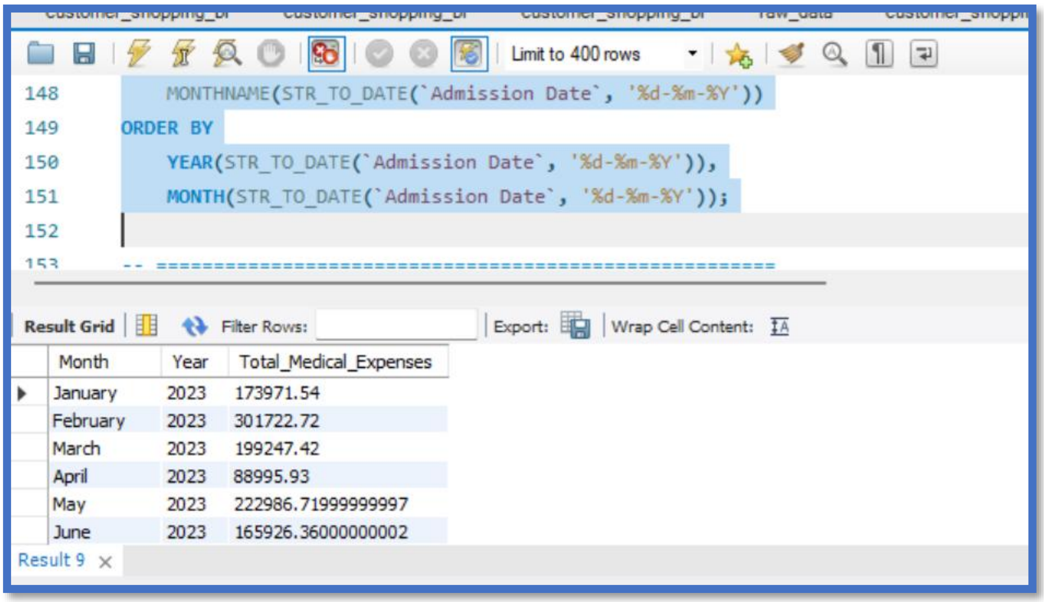
Objective

Calculate the total medical expenses for each month.

SQL Query

```
SELECT
    MONTHNAME(STR_TO_DATE(`Admission Date`,'%d-%m-%Y')) AS Month,
    YEAR(STR_TO_DATE(`Admission Date`,'%d-%m-%Y')) AS Year,
    SUM(`Medical Expenses`) AS Total_Medical_Expenses
FROM hospital_data
GROUP BY
    YEAR(STR_TO_DATE(`Admission Date`,'%d-%m-%Y')),
    MONTH(STR_TO_DATE(`Admission Date`,'%d-%m-%Y')),
    MONTHNAME(STR_TO_DATE(`Admission Date`,'%d-%m-%Y'))
ORDER BY
    YEAR(STR_TO_DATE(`Admission Date`,'%d-%m-%Y')),
    MONTH(STR_TO_DATE(`Admission Date`,'%d-%m-%Y'));
```

Output



Observation

The query generates a month-wise report of the total medical expenses.

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8. SQL Concepts Used

- SELECT Statement
- Aggregate Functions (SUM, AVG)
- GROUP BY
- ORDER BY
- LIMIT
- Date Functions (DATEDIFF, STR_TO_DATE, MONTHNAME, YEAR)
- Data Aggregation
- Business-Oriented SQL Analysis

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9. Key Insights

After executing all SQL queries, the following insights can be derived:

- The dataset provides a comprehensive overview of hospital operations.
- Patient distribution varies across different departments.
- Some hospitals incur significantly higher medical expenses than others.
- Average hospital stay differs by department.
- City-wise patient counts help identify areas with higher healthcare demand.

- Monthly expense reports can be used to monitor healthcare spending trends.

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10. Conclusion

This project demonstrates the practical application of SQL for healthcare data analysis. By using aggregate functions, grouping, sorting, and date functions, meaningful insights were extracted regarding patient volume, doctor availability, departmental performance, hospital stay duration, and medical expenses. The project strengthened SQL querying skills and provided hands-on experience with real-world healthcare data, making it a valuable addition to a data analyst portfolio.

References

- Dataset: Hospital_Data.csv
- Database: MySQL
- Tool: MySQL Workbench